Review Article

A Qualitative Systematic Review of Industrial Corrosive Substances in India Using the PRISMA Framework

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Abstract - This study compares and analyses the regulation, supply chain and usage of industrial corrosive substances in India, with particular focus on their accessibility, control and safe handling across the different stages of the supply chain. Despite the development of regulatory and safety frameworks, the reviewed literature indicates gaps in the extent to which these controls are implemented across manufacturing, transportation, storage, purchasing and end-use. This review aims to review the regulatory and industrial frameworks governing hazardous materials, to analyse the existing evidence on the movement and accessibility of corrosive substances, and to determine the regulatory, operational and monitoring challenges affecting their effective control. A qualitative systematic review approach is used to adhere to the PRISMA methodology. After screening, a total of 186 studies were identified from the Scopus database and analysed by thematic synthesis, which were further categorised into licensing, traceability, buyer verification and intended purpose of purchase, quantity control and recording, regulatory inspection, storage and safe-storage compliance, violations and penalties, ease of purchasing and market accessibility, and usage and end-use patterns. The findings show that, although regulatory and safety measures have developed significantly, the reviewed literature reported a mismatch between the existence of regulatory requirements and their implementation in practice. While licensing, purchaser identification, recording, inspection and safety-management requirements are present, their effectiveness varies across industries and settings owing to limitations in monitoring, traceability, organisational capacity and enforcement. The study shows that regulatory frameworks, institutional capacity and accessibility of corrosive substances play a significant role in shaping chemical safety in India. It highlights the importance of integrated supply-chain control, improved monitoring and stronger accountability mechanisms for the safe management of industrial corrosive substances.

Keywords: hazardous materials, industrial corrosive substances, chemical safety, supply chain, regulatory control

1. Introduction

Study on hazardous materials and industrial corrosive substances has become a focal point of research in India, as it plays a key role in industrial production, transportation, storage and commercial activities while at the same time it can create significant risks to workers, communities and the environment. Since the expansion of industrialisation in India, there have been various measures towards chemical safety, beginning with industrial safety legislation and followed by various necessary regulatory and preventive measures for the handling of hazardous substances (Sivaprakash et al., 2015). There are minimum safety requirements prescribed for the transportation and handling of hazardous chemicals according to regulatory and industrial safety practices. These requirements are embedded in the larger system of hazardous chemical management, which includes the need for safe transportation, appropriate handling, among others in order to prevent accidents during the movement of chemicals (Sivaprakash et al., 2015). The regulation of acid and other corrosive substances has also received judicial and legislative attention, with requirements

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relating to licensing, identification of purchasers, recording of sales and regulation of the possession and sale of acids being some factors that have been monitored (Kuriakose et al., 2017). Hazardous chemicals are also used across major industrial activities and can thus also create risks during production, storage, transportation, among other processes. With chemical accidents having the potential to majorly impact workers as well as populations living around industrial areas (Renjith & Madhu, 2010). Through accidental or intentional exposure, corrosive substances can also be associated with serious health consequences, with the availability of acids being a fundamentalt concern in relation to acid violence and public safety (Kuriakose et al., 2017). These efforts thus highlight the importance of controlling hazardous materials while keeping in mind their significance for industrial safety, public health and also importantly environmental protection, particularly where hazardous substances are transported, stored and used across different settings.

While progress has been observed in this area, there still remains a major need for the adressal of the issues of effectively having control over hazardous materials across different stages of the supply chain, such as manufacturing, transportation, storage, purchasing and end-use. While, the understanding of why the effectiveness of these controls differs so widely across industries, locations and importantly, stages of the supply chain, especially for corrosive substances, is limited. Thus, this studyaims to systematically address this gap by reviewing the regulation, supply chain, accessibility, handling and usage of hazardous and industrial corrosive substances, particularly in India.

2. Importance of the Study

The existing literature shows that there exists a lack of integrated knowledge around the regulation, handling, accessibility and movement of hazardous and corrosive substances across different stages of the supply chain in India. Studies often examining industrial hazards, chemical accidents, environmental regulation or acid violence, focus on them as separate areas rather than as interconnected parts that affect one another (Sharma et al., 2014; Rajeev et al., 2019; Roy & Kshirsagar, 2021). Some studies focus on the legal and environmental framework for managing hazardous chemicals while others point towards gaps in risk management, accident prevention and preparedness and the need for more integrated and scientifically informed approaches to chemical risk management (Sharma et al., 2014; Roy & Kshirsagar, 2021). The impact of these gaps are chemical exposure, occupational injuries, public health risks, environmental damage, in addition to the continued vulnerability of populations located around hazardous industrial and storage facilities (Rajeev et al., 2019; Choudhary et al., 2026). The conceptual framework of this review is built on the principle of chemical risk management and supply chain safety as a multidimensional concept, which includes regulatory control, traceability, verification of purchasers, quantity control, inspection, storage, enforcement and the actual use and end-use of corrosive substances (Sharma et al., 2014; Roy & Kshirsagar, 2021). The framework, in addition to shedding light on the interlinkages between regulation, supply chain practices, accessibility and safety outcomes, also helps offer a foundation for analysing the movement and use of industrial corrosive substances at the national level. This systematic review, thus aims to examine the regulation, supply chain and usage of hazardous materials and industrial corrosive substances, particularly in India so as to help critically analyse the existing evidence around their accessibility, control and safe handling. This study brings together and translates various empirical evidence, health studies, industrial safety research and regulatory analysis to address the gaps in current knowledge and provide empirical evidence for improving the control and safe management of corrosive substances in India. The review uses a thorough and systematic approach that integrates qualitative analyses of peer-reviewed research. The inclusion criteria are geared towards Indian literature covering regulatory,

industrial, health, supply chain and usage related evidence, giving a nationwide perspective. Findings are thematic and cover on licensing, traceability, buyer verification and intended purpose of purchase, quantity control and recording, regulatory inspection, storage and safe-storage compliance, violations and penalties, ease of purchasing / market accessibility and usage and end-use patterns. Over the years, hazardous chemical management in India has undergone a lot of development through environmental, industrial and safety regulations, while hazardous substances continue to be used across manufacturing, storage, transportation and other commercial activities (Sharma et al., 2014; Rajeev et al., 2019). These measures cover several areas, such as industrial safety, chemical risk management, storage and emergency preparedness, reflecting the State's efforts to control the risks associated with hazardous materials in different settings. The significance of this study is in its attempt to develop an integrated and comprehensive view about the functioning of these controls in practice, and the way these controls address the risks created by the availability and movement of corrosive substances.

This study systematically synthesizes evidence across regulatory control, supply chain accessibility, industrial handling, health outcomes and end-use patterns of hazardous and corrosive substances in India in a single PRISMA-based framework. The review brings together evidence addressing areas such as chemical accidents, environmental regulation, industrial risk, acid violence, regulatory control and end-use, thereby providing an integrated perspective on the movement, accessibility, control and use of corrosive substances in India. This review's integrated scope differs from previous chemical safety and health based studies, which tend to examine individual incidents, specific industrial settings or the consequences of exposure without bringing together the complete chain through which corrosive substances are supplied, purchased, stored and used. This study critically analyses the previous literature on hazardous materials and industrial corrosive substances in India covering regulation, supply chain control, accessibility, handling, health outcomes and usage. In this review, the industrial corrosive substances refer to the corrosive chemicals that are produced, supplied, stored, transported or used in the industrial and commercial supply chains, which includes substances that may subsequently enter household, medical or other end-use settings. The evidence base therefore includes direct industrial evidence, such as studies of workplace exposure, chemical safety, storage, transportation and industrial regulation, as well as including broader public-health and accessibility evidence, and including the hospital poisoning cases, household exposure and acid violence. The latter evidence is also used to examine the consequences and patterns of the accessibility and end use of corrosive substances and to help provide context for assessing how contnrols on the supply-chain operate beyond the industrial setting. The study employs a qualitative systematic review methodology using the PRISMA framework which involves synthesising the literature to assess regulatory frameworks, supply chain practices and safety measures related to corrosive substances in a range of industrial, commercial, medical and other usage contexts. One of the key aims is to understand the effectiveness of these controls in achieving their safety objectives and to highlight implementation gaps, weaknesses in monitoring, difficulties in traceability and risks associated with easy access to corrosive substances. Accordingly, the review addresses three questions: (1) What regulatory and supply-chain controls govern the licensing, verification, recording, inspection and storage of the industrial corrosive substances within India? (2) What does the existing literature help indicate about the accessibility, purchasing and end-use patterns of these corrosive substances across different settings? (3) What implementation gaps and challenges are reported in the monitoring, traceability, enforcement and safe management related to these substances? Studies of major accident hazard industries have shown that hazardous substances can create risks during production, transportation and storage, while risk assessment research in India has also pointed towards the need for clearer and more systematic approaches to controlling major industrial hazards (Rajeev et al., 2019; Roy &

Kshirsagar, 2021). The recent investigation of a corrosive chemical leak at the Tughlakabad Inland Container Depot further shows how failures in chemical spill management and risk communication can contribute to illness among exposed populations (Choudhary et al., 2026). At the same time, studies on acid violence show that corrosive substances also have a distinct public safety dimension when they are obtained and used for intentional harm, with the consequences extending beyond the initial physical injury to psychological, social and legal difficulties for survivors (Ghosal & Chattopadhyay, 2021; Thussu, 2025). However, there is a persistent gap between the existence of controls and the ability to examine how these controls operate across the full supply chain. This study's integrated approach brings fresh insights to the literature on hazardous materials regarding the structural and regulatory conditions affecting the safe movement and use of corrosive substances in India. It provides useful policy, academic and practice implications to inform the design of more effective regulatory, monitoring and supply chain interventions. Future studies could broaden their focus beyond individual database reviews to include multiple databases, government inspection records, chemical accident databases, regulatory enforcement records and industry level procurement and distribution data so as to gain a fuller picture of the movement, accessibility and use of hazardous corrosive substances in India.

3. Materials and Methods

The method used for the systematic review was adopted from the guidelines on the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) (Moher et.al. 2015; Page, M.J. et.a., 2021). To locate, select, and study pertinent material, as well as to extract and define recommendations for future research, a traditional thematic systematic review synthesises on existing research and analyses explicit methodologies employed in previous paper (Paul, J & Dhiman, R. 2021). Thematic synthesis was conducted in three steps: line-by-line coding of the reported findings, development of descriptive themes, and generation of final analytical categories. Each of the 186 included studies was therefore reviewed and coded according to the substantive issues that were being addressed in the study. Initial codes were grouped according to similarities in the subject matter and relevance to the objectives of the review, and these descriptive themes were subsequently also refined into the nine final analytical categories of licensing, traceability, buyer verification and intended purpose of purchase, quantity control and recording, regulatory inspection, storage and safe-storage compliance, violations and penalties, ease of purchasing / market accessibility, and usage and end-use patterns. Where a study addressed more than one aspect, relevant findings were assigned to more than one analytical category.

The coding framework used to organise the included studies and develop the final analytical categories is presented in Table 1.

Due to the diversity of study design, outcome measures and regulatory context in the literature included, a qualitative synthesis was used instead of a meta-analysis. Given the diversity of study designs included in the review, a formal risk-of-bias assessment utilising a single standardized appraisal tool was not undertaken. The included studies thus comprised clinical studies, case reports, regulatory and legal analyses, and industrial-safety research with different methodological approaches and outcome measures. Instead, the evidence was considered in relation to the study design, methodological clarity, relevance to the review objectives and also particularly the limitations reported by the respective authors. These considerations were taken into account when interpreting and synthesising the findings, particularly where evidence was derived from the individual case reports or studies with limited methodological detail. The rationale for systematic review planning is to provide a clear and transparent process for identifying,

selecting, and analysing existing research (Moher et.al. 2015). SLR can be classified into three categories: theory-based, method-based, and domain-based (Paul & Dhiman, 2021; Paul & Criado, 2020). In this study, a traditional thematic systematic review was used to study hazardous materials and industrial corrosive substances in India with particular focus on their regulation, supply chain, accessibility, handling, and usage.

Analytical category	Coding focus
Licensing	Licensing requirements and regulatory permissions
Traceability	Records, tracking and documentation of substances
Buyer verification and intended purpose of purchase	Purchaser identification and stated purpose of use
Quantity control and recording	Quantity restrictions, purchase records and transaction documentation
Regulatory inspection	Inspection, monitoring and enforcement mechanisms
Storage and safe-storage compliance	Storage practices and safety requirements
Violations and penalties	Non-compliance, offences and associated penalties
Ease of purchasing / market accessibility	Availability, accessibility and ease of obtaining corrosive substances
Usage and end-use patterns	Industrial, occupational, household, accidental and intentional uses

Table 1.: Coding framework for the thematic synthesis of included studies

The Scopus database was searched using the search strategy: acid* OR "corrosive substance*" OR "corrosive chemical*" OR "corrosive material*" OR "corrosive agent*" OR "industrial acid*" OR "strong acid*") AND (India OR Indian) AND ("acid attack*" OR "acid assault*" OR "acid violence" OR "acid throwing" OR "acid burn*" OR "chemical assault*" OR "chemical violence" OR licens* OR permit* OR regulation* OR legislation OR compliance OR traceab* OR tracking OR "buyer verification" OR "purchaser verification" OR "end user verification" OR "intended use" OR "purpose of purchase" OR "quantity control" OR "quantity restriction*" OR inspection* OR audit* OR monitoring OR storage OR "safe storage" OR violation* OR offence* OR offense* OR penalty OR penalties OR sanction* OR prosecution OR enforcement OR procurement OR purchasing OR supplier* OR distribut* OR retailer* OR dealer* OR injur* OR victim* OR fatalit* OR mortalit* OR burn* OR trauma OR morbidity OR "clinical outcome*" OR "health outcome*".

The inclusion criteria consisted of peer-reviewed English-language publications that were focused on the substantive aspects of hazardous materials and industrial corrosive substances, including their regulation, supply chain, accessibility, handling, health outcomes and usage within India. Studies that did not address

these aspects in the Indian context were thus excluded. Following the application of the subject-area and publication filters, 970 records remained for screening. The records were further also assessed according to language, document type, journal source, publication stage and open-access availability. Following title and abstract screening and subsequent full-text assessment, 784 records were excluded for not meeting the predefined eligibility criteria, resulting in 186 studies being included in the qualitative synthesis. The screening and selection process is presented in the PRISMA flow diagram in Figure 1, with the numbers of the records excluded at each stage corresponding to the reported screening and eligibility process.

The papers were then analysed according to the nine analytical categories used in this review: licensing, traceability, buyer verification and intended purpose of purchase, quantity control and recording, regulatory inspection, storage and safe-storage compliance, violations and penalties, ease of purchasing / market accessibility, and usage and end-use patterns.

The characteristics of the included literature and its distribution across the nine analytical categories are summarised in Table 2.

4. Literature Review

Hazardous chemicals and corrosive substances can cause poisoning, burns, occupational exposure and accidental injury when they are improperly handled or stored, yet are essential in industrial, commercial, laboratory and household uses. Mannan et al. (2007), in a review conducted of 24 studies pertaining to chemical assault, examined 771 cases reported over a period of 40 years and as a result found that chemical substances were being used as weapons in different social and geographical settings. The authors importantly found that the ease with which these chemical agents could be obtained contributed to their use in assaults, and further also discussed the types of agents used, the circumstances of the attacks and importantly the role of legislation in controlling access to corrosive substances.

Poisoning studies in India have also dealt with the effects on health related to exposure to hazardous chemicals. Dhara and Dhara (2002), reviewed the health effects of the 1984 Bhopal gas disaster, particularly the consequences of exposure to methyl isocyanate among the affected population, and reported the acute effects on the eyes and respiratory system and later even identified persistent ocular and respiratory problems, in addition to neurological, reproductive, neurobehavioral and psychological effects. Importantly, the review recommended long-term monitoring and better exposure assessment in studies of chemical disasters due to current limitations in exposure classification.

Kohli et al. (2008) examined childhood poisoning cases that were admitted to a tertiary care centre in North India, and found that most cases were accidental, and that substances available within the household environment were the major cause of poisoning among children. Ramesha et al. (2009) similarly also examined acute poisoning cases in a tertiary care hospital in Karnataka and reported differences existing in outcome according to the substance involved. Importantly, corrosive poisoning was associated with a particularly high mortality in their study. These studies thus show that hazardous substance exposure can occur through deliberate misuse but may also be caused due to ordinary household and community environments.

Studies conducted in poison information centres have provided an even broader picture of the substances involved in poisoning. For instance, Singh et al. (2011), a study examining patients with acute toxicity admitted to an intensive care unit in an urban Indian hospital, reported that the clinical outcome varied with the substance and severity of exposure. Lakshmi et al. (2013), another study examining corrosive alimentary

injuries in an Indian hospital setting, described the clinical characteristics and also the complications associated with corrosive ingestion. Their study also discussed the continuing occurrence of corrosive injuries in India, and particularly the difficulties that were associated with managing severe injuries of the alimentary tract.

Ahuja et al. (2015) examined acute poisoning cases admitted to a tertiary intensive care unit in northern India. This study reported the clinical profile and outcomes of the patients, whereas Unnikrishnan et al. (2015) approached this issue from an occupational perspective and examined safety management practices in 30 small and medium enterprises in India. Their study considered the safety practices followed by these enterprises as well as the barriers affecting their implementation, and found that safety practices were influenced by factors such as awareness, resources and also the ability of smaller enterprises to implement formal safety systems.

Safety practices in laboratories have also been examined in the Indian context. Mustafa et al. (2008) being an example of the same. This study discussed laboratory safety in India and described the importance of appropriate handling procedures, protective equipment and safe working practices when dealing with hazardous materials. Their work thus importantly considered safety as a part of routine laboratory practice compared to it being considered as a response to accidents.

Vinodkumar and Bhasi (2009), using responses from 2,536 employees, focused on safety climate in eight major accident hazard chemical industrial units in Kerala. The study examined several dimensions of safety climate, and in doing so, found a relationship between safety climate factors and self-reported accidents. The findings reported indicate that organisational practices, employee perceptions and safety systems, all influence the likelihood of accidents in chemical industries. This study thus becomes particularly relevant to the handling of hazardous materials because it places chemical safety within the wider organisational environment and not treating accidents as separate events.

The regulation of hazardous industrial activity has also been examined, specifically in relation to monitoring and enforcement. Kathuria and Sterner (2006), in their study of chemical plants in Ankleshwar, Gujarat, examined a two-tier system in which an industry association acted as an intermediary between individual plants and government regulators. Their analysis considered monitoring, implementation of rules and enforcement and found that the system had partially satisfied the conditions required for effective collective regulation. The study importantly also identified and noted the difficulties faced by regulatory authorities in monitoring numerous small industrial units, and the role that industry associations could play in supplementing formal government regulation.

Sengupta et al. (2016) studied the risk assessment practices for industrial accidents in India. Evaluating an industrial accident risk assessment framework and considered the way industrial hazards could be assessed and managed, the study becomes relevant to chemical safety because risk assessment provides a basis for identifying hazardous activities before an accident occurs and for determining appropriate preventive measures.

Roy et al. (2017) examined children hospitalised with acute poisoning in New Delhi, and reported that medicines, corrosives, detergents and kerosene were among the most common substances involved in poisoning. The cases were predominantly accidental and occurred largely among younger children. All of these findings thus indicate the importance of the immediate environment in determining access to hazardous substances and also show that corrosive substances can be encountered in homes as well as in industrial settings.

Characteristic	Description
Study type	Industrial and occupational safety studies; regulatory and legal analyses; hospital and poisoning studies; case reports and case series; studies of chemical exposure and acid violence
Year range	No publication-year restriction was applied
Geographic setting	India, including national-level studies and studies conducted in specific states, cities and industrial settings
Licensing	Studies examining licensing requirements, legal restrictions and controls on the sale and possession of corrosive substances
Traceability	Studies addressing purchaser records, documentation and tracking of hazardous substances
Buyer verification and intended purpose of purchase	Evidence concerning purchaser identification and declaration or verification of intended use
Quantity control and recording	Evidence concerning quantity restrictions, transaction records and purchase documentation
Regulatory inspection	Studies addressing inspection, monitoring, safety systems and regulatory enforcement
Storage and safe-storage compliance	Studies examining storage conditions, workplace safety practices and compliance with hazardous-material requirements
Violations and penalties	Evidence concerning non-compliance, offences, enforcement and penalties
Ease of purchasing / market accessibility	Studies examining availability and accessibility of corrosive substances through commercial channels
Usage and end-use patterns	Evidence covering industrial, occupational, household, accidental, intentional and other end-use settings

Table 2.: Summary of the evidence base included in the review

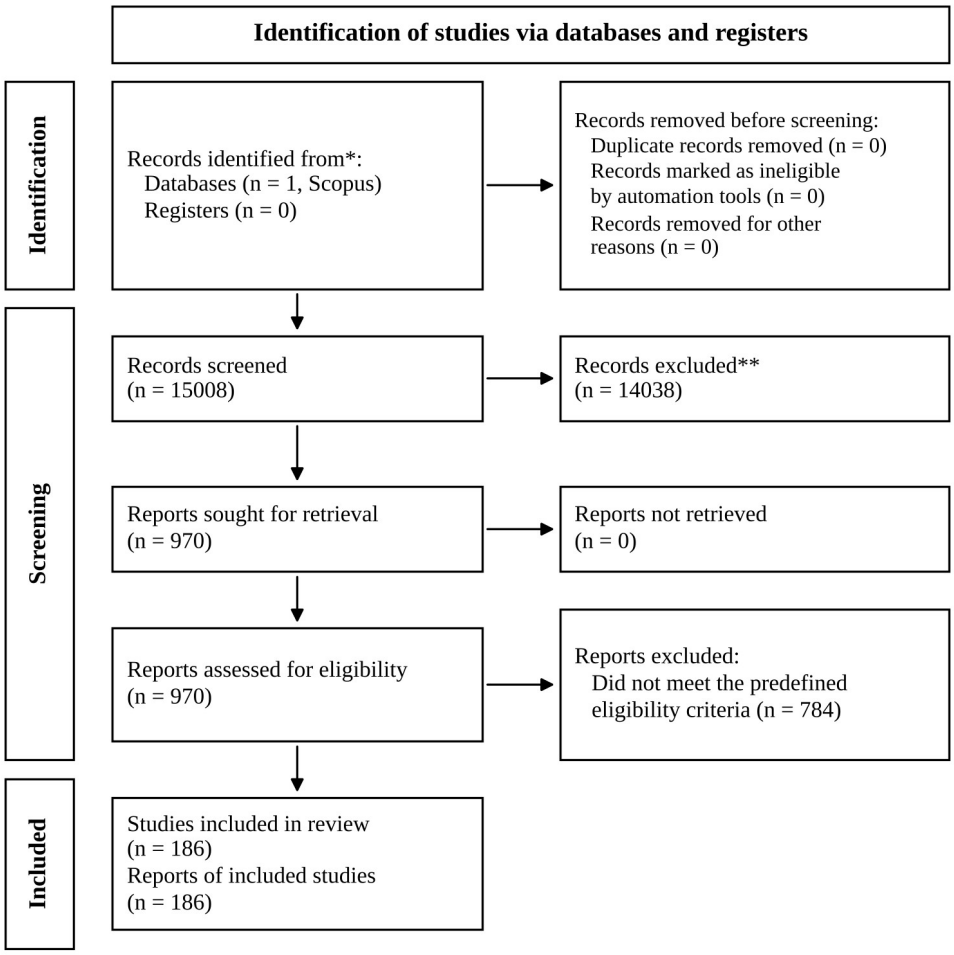


Fig. 1.: PRISMA Diagram

Patel (2014) examined acid attacks in India through content analysis of newspaper reports published between 2009 and 2013, and their study identified and examined the circumstances and methods associated with the attacks in 58 reported cases. The findings showcased that the incidents occurred among young women as well as in rural areas and involved different personal and social circumstances. Thus an indication of the ways in which corrosive substances can move from legitimate or ordinary use into deliberate harmful use is provided through this study.

Singh et al. (2018) examined 52 acid attack cases reported between July 2012 and June 2017 and found that most victims were young women and that many of the attacks were associated with relationship, marriage or personal disputes. Facial disfigurement was reported in all cases and ocular involvement was common, thus these findings demonstrate the severe consequences associated with the deliberate use of corrosive substances and provide a clinical as well as social context for the examining of the availability of such hazardous substances.

Thus, the legal treatment of acid and other corrosive substances has developed alongside these concerns. The Law Commission of India (2009), in Report No. 226, examined the inclusion of acid attacks as specific

offences in the Indian Penal Code and even considered compensation for victims. The report, reviewing cases of acid attacks in India, the consequences of such attacks and laws adopted in other countries, recommended specific legal provisions addressing acid attacks and thus considered the regulation of the offence alongside compensation and victim protection.

Kay (2013) reported on the Indian Supreme Court's efforts to restrict the sale of acid following the growing concern over acid attacks and even described measures concerning the sale and purchase of acid and the identification of purchasers, thus providing contemporary documentation of the movement towards regulating the retail sale of corrosive substances.

In relation to enforcement of these restrictions, the availability of corrosive substances through ordinary commercial channels has also been examined. The Delhi Commission for Women, in a market survey throughout Delhi, had representatives approach shops selling acid and examine whether restrictions concerning its sale were being followed. The survey was thus able to provide evidence that corrosive substances could still be obtained through ordinary retail outlets despite their being the existence of restrictions, and this issue was subsequently considered in future judicial proceedings concerning the implementation of the regulatory framework.

Importantly, the regulation of hazardous substances is not limited to acid used in assaults. Gupta, Saksena, and Baris (2019), in their examination of the environmental enforcement and compliance in India using data from large water- and air-polluting plants in Punjab, found that inspections by the regulatory authority were associated with plant-level compliance and therefore enforcement activity was directed towards frequent violators. These findings are able to demonstrate the role inspection and enforcement plays in influencing compliance with environmental regulations.

Pagdhune et al. (2020) conducted an analysis of poisoning cases reported to the Poison Information Centre in Ahmedabad over a three-year period, particularly examining the substances involved in reported poisoning cases and included household chemicals among the identified categories. The authors also noted that the chemical nature of a substantial proportion of reported cases was unknown. The study was able to thus illustrate the role of poison information centres in documenting exposure patterns and identifying the substances associated with poisoning in the community.

Chatterjee et al. (2020) examined acute poisoning cases in a tertiary care hospital in West Bengal and found that corrosive substances accounted for a measurable proportion of the poisoning cases, while accidental poisoning was common and many incidents occurred at home. The study also recorded deaths among the patients examined. These findings were able to reinforce the role of household access and accidental exposure in the occurrence of poisoning involving hazardous substances.

The regulatory dimension of chemical use has also been examined, particularly in relation to the availability of hazardous substances through commercial and digital channels. Government and judicial interventions have increasingly addressed the sale of acid, including requirements relating to the identification of purchasers and restrictions on unrestricted retail sale, and these measures are able to thus indicate a shift from treating corrosive substances only as products of legitimate industrial or commercial use towards considering the conditions under which their sale and possession should be controlled.

The clinical literature also continued to document serious consequences from corrosive exposure, with Chaurasia et al. (2024) examining four fatal cases of acute corrosive acid ingestion through autopsy

findings. Three of these cases involved hydrochloric acid while one involved sulphuric acid. The cases are able to demonstrate the potential for corrosive substances to cause fatal internal injury when ingested.

Krishnasamy et al. (2024) examined 2,123 poisoning cases in a tertiary care hospital in South India and assessed factors associated with poisoning-related mortality, and found that corrosive poisoning represented an important category of poisoning in the study, particularly among younger patients. The study also importantly examined the relationship between the type of poison, clinical condition and mortality, showing the continuing burden of corrosive exposure in hospital settings.

More recent research has also examined corrosive exposure in occupational settings. Das et al. (2026) reported a series of fatal formic acid poisoning cases, including both occupational and intentional exposure, among workers involved in rubber latex processing. The study thus demonstrated the serious consequences that can arise when corrosive substances are present in industrial workplaces.

The literature therefore covers several different circumstances in which hazardous and corrosive substances are encountered. While Hospital and poison information centre studies have documented accidental and intentional poisoning, occupational studies have examined particular safety practices and exposure in industrial environments.

Research on chemical industries has considered safety climate, risk assessment, monitoring and even regulatory enforcement. Studies of acid attacks have examined the deliberate misuse of corrosive substances and the consequences of their availability, while legal and government reports have addressed restrictions on their sale and possession. Together, these studies are able to provide the background for examining how hazardous corrosive substances move through legitimate channels of supply, how their handling is regulated and also how existing controls operate in practice, with industrial and regulatory studies also providing direct evidence on supply-chain controls and broader health.

5. **Analysis**

The analysis distinguishes between formal regulatory requirements and evidence of their implementation in practice. Regulatory and legal studies were used to identify the requirements established by the existing framework, while empirical and observational studies were considered for evidence concerning how these requirements operate in practice.

5.1 Licensing

The regulatory framework that governs hazardous materials in India places the licensing and regulatory control within the broader management of hazardous substances across their supply chains. Dadeech (2020) examined the requirements governing hazardous materials across manufacturing, storage and transportation and emphasized the role of regulatory controls in managing the risks associated with these substances in their assessment of the regulatory framework for hazardous materials in the Indian chemical industry. The study further places hazardous-material regulation within the complete supply chain instead of treating control as limited to the point of manufacture or sale. Chaturvedi (2018), specifically examining the sale of acid in India, described the legal restrictions introduced to regulate acid sales and the subsequent development of the rules governing possession and sale. The regulation of acid sales therefore forms a part of a wider statutory framework that is intended to control the access to substances capable of causing serious injury.

The judicial development of these controls is further discussed by Tandon (2015), which examines the Supreme Court's intervention in *Laxmi v. Union of India* and particularly its recognition of the easy availability of acid in markets. The judgment resulted in guidelines for regulating acid sales and even introduced requirements intended to place greater responsibility on those selling acid. Similarly, Kazerooni et al. (2020), in their systematic review of legislative responses to chemical assault, identified India as having introduced controls over the supply, sale and procurement of acid, including restrictions concerning the age of purchasers, identification requirements, declaration of purpose, seller licensing and maintenance of purchaser records. Ahmad (2011) also discusses the development of the Indian legal response to acid attacks and the movement from general criminal provisions towards specific regulation of acid sales. These studies collectively describe licensing as part of a regulatory structure that connects the seller's legal responsibility with the controlled sale of corrosive substances.

5.2 Traceability

Traceability of corrosive substances refers to the information available about the source and circumstances of exposure in poisoning studies. Swain et al. (2016) conducted a retrospective study of 64 fatal corrosive-ingestion cases from South and South-East Delhi between the years of 2005 and 2014, and found that sulphuric acid accounted for 68.75% of the cases while household cleaning substances accounted for 39.06%. The exact source of these corrosive substance was unknown in as much as 56.26% of the cases, while 87.5% of the deaths ended up being classified as suicidal. These findings are able to thus demonstrate that corrosive exposure can involve both concentrated acids and even commercially available household substances, with the source of the substance forming part of the information recorded during forensic investigation.

A similar pattern of source identification appears in the surveillance of poisoning. Srivastava et al. (2005), analysed 2,719 calls received by the National Poisons Information Centre at the All India Institute of Medical Sciences between April 1999 and March 2002, and classified poisoning exposures according to the substances involved, including household products and even industrial chemicals. The study was able to find that household products constituted 44.1% of the reported exposures, while industrial chemicals accounted for 7.9%. Similarly, Kumar et al. (2018), in their examination of ten years of calls to the same centre, analysed 16,420 poisoning-related calls, including 7,114 involving household products. In their study, household cleaners accounted for 21.8% of these household-product exposures, with 66.8% of household-product cases unintentional. These studies thus demonstrate that information concerning the substance and its reported source is also incorporated into poisoning surveillance, while the circumstances of acquisition and exposure are recorded alongside the clinical or forensic outcome.

5.3 Buyer Verification and Intended Purpose of Purchase

The regulation of acid sales also incorporates the requirements that relate to the identity and the stated purpose of the purchaser. Kumar (2021), conducted a socio-legal examination of acid attacks in India, and described the regulatory requirements introduced following judicial intervention. These requirements included restrictions related to the age of purchasers, production of identification, declaration of the intended purpose of purchase and further obligations placed upon sellers. The purchaser is therefore not anonymous, but is instead treated as an identifiable individual whose proposed use of the substance forms an important part of the transaction. The regulatory approach has thus consequently helped connect the sale of corrosive substances with verification of the person purchasing them and the stated reason for the purchase.

Calcini (2022), examined legal measures and motivation trends relating to acid attacks across India, Pakistan, Bangladesh, Colombia and Cambodia, and similarly placed the regulation of chemical supply and procurement within the preventive response to the acid violence. The Indian framework forms a part of this much wider approach which is a combination of restrictions on access, requirements concerning purachsers, and also the circumstances in which corrosive substances may be obtained. The studies therefore are able to present buyer identification and intended use as connected elements of the regulatory control of the acid sales.

5.4 Quantity Control and Recording

Control over the quantity of acid that is purchased is another important component of the regulatory framework. Calcini (2022) described the Indian regulatory response as extending even beyond the identification of the puracher and declaration of purpose, to the recording of the quantity of acid sold. Similarly, Kumar (2021) discussed the requirement for sellers to maintain the transaction-related information, including details concerning the purchaser and the also amount of acid involved. Thus, the quantity purchased and the intended use forming part of the impotant information associated with the buyer in such transactions. However, it is important to note that evidence concerning actual implementation of quantity monitoring or even recording is limited, despite the literature establishing the formal recording requirement.

5.5 Regulatory Inspection

Regulatory inspection has been closely related to the implementation of hazardous-material controls in the industrial settings. Gupta et al. (2019) examined environmental enforcement in India using the data from 117 large water-polluting and 109 large air-polluting plants in Punjab, and found a relationship between regulatory inspection and environmental compliance. The study is able to thus demonstrates the role of inspection in ensuring the translation of regulatory requirements into operational compliance within regulated industrial facilities. Kathuria and Sterner (2006) examined the chemical-industrial cluster at Ankleshwar, also found that regulatory monitoring operated within conditions of limited regulatory resources and information asymmetry. Further, their analysis of monitoring, enforcement and fines demonstrates how inspection forms part of the mechanism through which regulated industries are brought into compliance.

Sengupta et al. (2016), in their evaluation of the risk-assessment framework for industrial accidents in India, examined the institutional and regulatory arrangements that govern the industrial accident prevention and were able to identify weaknesses in the integration of risk assessment within the regulatory system. Vijalapura (2019) studied 1,029 employees across 22 major accident hazard industries in Karnataka, and found variations in the safety climate and safety-system factors across the organisations that were under study, with safety-system characteristics being associated to organisational attributes. These findings are able to place inspection and regulatory oversight within a wider system of organisational safety management.

A direct example that involves a corrosive chemical is provided by Choudhary et al. (2026), who investigated a corrosive chemical (2-chloro-5-chloromethylpyridine) leak from a container depot in Tughlakabad, Delhi, that affected school students. The investigation recorded 619 affected individuals, 92% of whom were female, with a median age of 13 years. The investigation was able to identify the absence of adequate standard operating procedures for verification of hazardous containers, poor implementation of

basic safety measures, the lack of a separate area for hazardous materials, and also inadequate spill management and deficiencies in risk communication. This incident demonstrates further how inspection and operational oversight extends beyond the existence of regulatory requirements and to the manner in which hazardous-material controls are implemented at facilities that handle such substances.

5.6 Storage and Safe-Storage Compliance

Safe storage and also handling of corrosive substances has been examined in laboratory, industrial and small-enterprise settings in India. Mustafa et al. (2008), conducted a cross-sectional study of clinical laboratories in India, and found that 78% of laboratories followed the hazardous-material regulations, while 95.6% of staff in the laboratories had been instructed in the safe handling of acids. Volatile and flammable chemicals were stored in designated areas in 73.95% of laboratories, and 94% of those designated areas were also suitably ventilated. The study also found that laboratories with accreditation were able to demonstrate much better safety parameters, thus helping link organisational systems with the implementation of safety practices in laboratories.

Unnikrishnan et al. (2015) studied 30 small and medium enterprises in and around Mumbai and other parts of India, examining the safety practices adopted within these facilities. The study was able to find that safety-management practices were inadequate in most of the SMEs studied and even identified financial constraints, awareness and organisational factors among the various issues that influenced safety practices. These findings demonstrate the influence that organisational capacity plays in the adoption of safety measures in workplaces that handle potentially hazardous materials.

Vinodkumar and Bhasi (2011) examined safety management in eight major-accident-hazard chemical companies in Kerala, involving 1,566 workers. The study found that employees in organisations that were certified under OHSAS 18001 consistently reported significantly higher levels of several safety-management practices than employees working in organisations without such certification, while safety rules and procedures emerged as a crucial predictor of safety behaviour across the groups. The findings are able to thus connect formal management systems with workplace safety practices adopted in chemical industries.

The consequences of inadequate control during storage and transfer are illustrated by Gangopadhyay and Das (2008), which examined an overflow that occured from a fuming sulphuric acid storage tank during unloading. The study describes the highly corrosive and reactive nature of oleum and further discusses the response to the storage-tank overflow, utilising the incident to identify engineering and operational safeguards for the handling of concentrated corrosive substances. Similarly, Sivaraman et al. (2022), analysing the Vizag styrene release, examined an accident that originated from an atmospheric storage tank and were able to associate the incident with deficiencies in safety management and operational preparedness. Although styrene is not a corrosive substance, the study provides important evidence from the Indian hazardous-chemical sector that importantly concerns the relationship between storage systems, operational safety and major chemical incidents.

5.7 Violations, Penalties and Enforcement

The criminal and judicial records surrounding acid attacks also reflect the consequences of misusing corrosive-substance. Khan et al. (2024), used retrospective data from the National Crime Records Bureau for 2017–2021, to examine the spectrum of acid attacks that occurred in India and were able to reporte a substantial number of cases progressing through the criminal justice system with varying levels of

pendency. The study placed acid violence within the broader functioning of police investigation, prosecution and even judicial proceedings. Similarly, Khan et al. (2026), examined acid attacks against women utilising NCRB data from 2017–2022 along with metropolitan police and trial data, and examined the progression of cases through the criminal justice system.

Kaur and Byard (2024), examined ten cases of acid attacks that involved children and adolescents in India, and were able to further demonstrate the consequences that are associated with the use of corrosive substances against vulnerable victims. The case series records the circumstances and forensic characteristics of these attacks and places the consequences of acid violence within medico-legal investigation. Together, these studies were able to show that enforcement of acid-related offences extends beyond the regulation of the substance itself, and to investigation, prosecution and the judicial handling of incidents after an attack has occurred.

5.8 Ease of Purchasing / Market Accessibility

Studies of acid exposure and violence document the availability of corrosive substances through ordinary commercial channels. Behera et al. (2016), for instance, reported a fatal accidental poisoning in an adult who had consumed sulphuric acid that had been marketed in a water bottle as a toilet-cleaning substance. The substance had been purchased from a local vendor and was subsequently stored together with drinking-water bottles. This case thus demonstrates how a corrosive substance sold through an ordinary local commercial setting can enter a household environment in a form that could create a risk of accidental ingestion.

Biswas and Das Chatterjee (2021), examined factors associated with acid attacks in West Bengal, and were able to identify perceived easy availability of acid in the open market as one of the important factors that were associated with acid attacks in the population studied. The study was able to thus connect the perceived accessibility of acid in ordinary markets with its subsequent use in violence. Similarly, Ahmad (2012) in their examination of the Indian legal response to acid attacks, discuss the accessibility of acid and the regulatory measures that were introduced to control its sale. The study links restrictions on acid sales with the wider legal response to acid violence, and places market accessibility within the preventive framework that has been developed in India.

5.9 Usage and End-Use Patterns

The end-use of corrosive substances in India has been documented across household exposure, intentional ingestion, accidental poisoning and also deliberate violence. Gupta et al. (2003), analysed 2,720 calls received by the National Poisons Information Centre over a three-year period, and were able to identify 995 childhood poisoning cases. Household products, including phenyl, detergents and corrosive substances accounted for 47% of these cases. These findings thus demonstrate that corrosive products form part of the household chemical environment to which the children are exposed.

Suganthi and Raja (2019), studied 106 poisoning patients at a tertiary care centre in Kancheepuram district, Tamil Nadu, and reported that 86.8% of the poisoning cases were suicidal in nature and that corrosive poisoning accounted for 17.9% of the cases. Family-related problems were reported among 63.4% of the patients. These findings connect the wider pattern of intentional poisoning observed in the clinical setting to corrosive ingestion.

Halesh et al. (2023), studied household hazardous-material exposure in rural Karnataka, particularly examining accidents and poisoning involving household HAZMAT and also documented the role of hazardous substances within domestic environments. The study was able to extend the understanding of end-use beyond deliberate poisoning by demonstrating exposure arising from the storage and use of hazardous materials within households. Salmani et al. (2024), through a multiple-case study that involved 22 acid-attack survivors, examined the motivations that were associated with acid attacks in India and were able to identify motives including jealousy, anger, greed and ego. The study consequently was able to place corrosive substances within two distinct end-use settings: routine household and occupational use on one hand, and deliberate interpersonal violence on the other.

Across these studies, corrosive substances appears in a range of end-use contexts that ranges from household cleaning and accidental exposure to intentional ingestion and deliberate assault. The nature of the end use is able to consequently determine the pathway through which the substance becomes harmful, while the regulatory controls surrounding purchase, handling and storage operate before these outcomes even occur.

An integrated summary of the nine analytical categories, their regulatory context, evidence of implementation, principal gaps and practical implications is presented in Table 3.

6. Limitations

This study has only retrieved articles from the Scopus database. Furthermore, a formal risk-of-bias assessment was not undertaken due to the substantial heterogeneity in the study designs and types of evidence included in the review, which should be considered when interpreting the findings. Future researchers can include more databases, government reports and industry reports in their review from Web of Science, Cambridge Core etc. Furthermore, the review was limited to only the English language, peer-reviewed publications, potentially overlooking relevant grey literature, government regulatory reports, industrial safety reports, and regional-language studies. As the regulation and handling of hazardous materials in India is changing rapidly, some very recent developments, regulatory amendments or even changes in the supply and distribution of corrosive substances may not be captured in the literature included. In addition to this, several included studies were only conducted in specific hospitals, states, industrial settings or case series, and their findings thus may not be directly generalisable to all regions or sectors in India. Accordingly, findings from geographically or sectorally specific studies are only interpreted within the context of the populations and settings examined.

7. Conclusion

A comprehensive qualitative systematic review on hazardous materials and corrosive substances in the industrial setting in India was carried out in this study, with a specific focus on licensing, traceability, buyer verification and intended purpose of purchase, quantity control and recording, regulatory inspection, storage and safe-storage compliance, violations and penalties, ease of purchasing / market accessibility and usage and end-use patterns, where there was found to exist limited amount of literature. The PRISMA 2020 checklist was utilised to help review the papers, and a total of 186 papers were able to be identified and retrieved under these nine factors under focus. The number of publications on the subject increased significantly over time, with earlier publications primarily focusing on industrial safety, chemical accidents and health outcomes that relate to hazardous chemical exposure, while the growth of studies on chemical risk assessment, hazardous chemical transportation, industrial safety and even regulatory control was

witnessed over time. In recent years, the number of publications has increased sharply, contributed by the growing attention towards chemical safety, regulatory compliance, chemical accidents and importantly the consequences of exposure to corrosive substance. The most important conclusion of this review is that there exist gaps within the controlling of hazardous materials across the entire supply chain, and that regulatory requirements for licensing, purchasing, storage and handling do not always translate into effective control over corrosive substances due to there being gaps in monitoring, traceability, inspection and also poor accountability systems. There is also a gap in policy and research as certain aspects such as verification of buyers, intended purpose of purchase, quantity control, legitimate accessibility and end-use of corrosive substances continue to be inadequately examined and thus found to be largely unrepresented within the scholarly record.

Future studies should therefore include a search of multiple databases, government regulatory reports, safety records in industries and even longitudinal studies so that a comprehensive understanding of the supply chain and the usage of hazardous materials can take place.

In addition to this, policy makers should move from the current substance-centric to a more supply-chain-centric regulation, with there being structural safeguards placed in addition to effective monitoring and accountability for creating meaningful and sustainable chemical safety in India. This is also supported by the evidence that has been synthesized across the review themes, particularly those concerning traceability, regulatory inspection, buyer verification, quantity control, violations and penalties, and ease of purchasing and market accessibility.

8. **Conflicts of Interest**

The author declares that there is no conflict of interest regarding the publication of this paper.

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Appendix

Table 3.: Integrative synthesis of the nine analytical categories

Analytical category	Formal control	Evidence of implementation	Principal gap	Practical implication
Licensing	Licensing and regulatory requirements for controlled substances	Evidence of regulatory controls across relevant settings	Variation in regulatory oversight and enforcement	Strengthen licensing oversight
Traceability	Documentation and records concerning substances and transactions	Source and transaction information documented in some settings	Limited traceability of substances in some cases	Improve documentation and tracking
Buyer verification and intended purpose of purchase	Purchaser identification and declaration of intended use	Requirements are documented within the regulatory framework	Limited evidence on consistent verification in practice	Strengthen purchaser verification
Quantity control and recording	Requirements for recording quantities and transactions	Limited evidence of actual quantity monitoring	Implementation evidence remains limited	Improve quantity recording and monitoring
Regulatory inspection	Inspection, monitoring and enforcement mechanisms	Inspection has been associated with compliance in some industrial settings	Resource and monitoring limitations	Strengthen inspection capacity
Storage and safe-storage compliance	Requirements for safe storage and handling	Evidence of storage and workplace safety practices in specific settings	Compliance varies across settings	Improve storage monitoring and safety practices
Violations and penalties	Legal provisions concerning violations and penalties	Evidence of enforcement and regulatory responses	Gaps in consistent enforcement	Strengthen accountability and enforcement
Ease of purchasing / market accessibility	Restrictions and controls on access and sale	Evidence of continued accessibility through some channels	Accessibility may persist despite formal restrictions	Improve control over points of sale and distribution
Usage and end-use patterns	Controls governing legitimate use and handling	Evidence across industrial, occupational, household and other settings	Evidence remains limited and context-specific	Improve monitoring of end-use patterns